

MUHAMMAD AHMED IMTIAZ

Junior AI/ML Engineer

ahmedimtqureshi@gmail.com | LinkedIn | Github | +92-336 5014829

Summary

Junior AI/ML Engineer with a strong foundation in machine learning, deep learning, and applied AI. Hands-on experience in building and deploying ML prototypes and production-ready components using Python, PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, XGBoost, and FAISS/RAG-based retrieval systems. Completed internships and a mentor-led DL/ML fellowship, worked on projects including voice-based AI agents, graph-based recommendations, and hybrid AI tools for smart farming. Familiar with REST API development using FastAPI and Flask, data preprocessing, feature engineering, model evaluation (AUC/F1), and reproducible ML pipelines. Seeking a junior role eager to learn and contribute to ML model development, deployment, and performance monitoring.

Technical Skills

- **Primary / Confident:** Python · SQL · Git / GitHub · PyTorch · scikit-learn · FastAPI · Flask · Deep Learning · CNNs (EfficientNet) · Transformers (BERT) · Embeddings · Model evaluation (AUC / F1) · Experiment tracking · Reproducible notebooks · MySQL · OpenCV · ngrok · Docker · Docker Compose · PostgreSQL · Apache Kafka · Go (Golang) · Next.js · Microservices · REST API design · InsightFace · PaddleOCR · Claude Code · Cursor · Windsurf
- **Familiar / Working knowledge:** TensorFlow · Hugging Face · FAISS · RAG · XGBoost · Rasa · n8n · Open-source STT/TTS (whisper.cpp, Vosk, Coqui/Piper) · C++ · Java · Gin framework · GORM · Kafka-Go · React
- **Professional Skills:** Problem-solving · Analytical thinking · Technical communication · Team collaboration · Time management · Adaptability

Experience

AI/ML Intern, Neuronix Technologies

Mar 2026 – Present

- Built a production-ready multi-tenant biometric identity verification platform integrating face matching (InsightFace), liveness detection (FSFM ViT-B16), OCR (PaddleOCR), and fingerprint matching (DeepPrint) as independent FastAPI microservices.
- Designed an event-driven pipeline using Apache Kafka for async result streaming, with a Go consumer persisting structured results to PostgreSQL and a Go/Gin admin backend serving a Next.js dashboard.
- Developed a data acquisition service (Go) and customer-facing portal (Next.js) as single entry points, coordinating parallel AI service calls and storing biometric evidence with full audit trail.
- Containerized and deployed the full 13-service stack using Docker Compose on a Linux VM, resolving production issues including Kafka networking, model memory constraints, and service orchestration.

AI/ML Intern & Team Lead, Codecelix NASTP

Nov 2025 – Feb 2026

- Working as AI/ML Intern and Team Lead, contributing to applied ML prototypes using free and open-source tools and coordinating technical tasks within the team.

- **Voice Customer Support Agent:** Contributed to a low-latency voice-to-voice pipeline (speech → text → response generation → speech) using open-source STT, LLM, and TTS tools; focused on low-latency, modular design, and clean documentation.
- **EchoSee Features:** Developed AI-based safety and accessibility features including sound-event detection , 360° directional audio awareness, and a real-time lip-reading prototype for caption generation.
- Documented architecture, tool selection, setup, and evaluation metrics; delivered runnable prototypes and demo videos following open-source and reproducibility best practices.

AI Intern, Dam Techhub

July 2025 – August 2025

- Cleaned and preprocessed Amazon co-purchase graph (edges + node features).
- Unsupervised Node2Vec embeddings (64-dim) and supervised GraphSAGE link-prediction model with improved AUC (0.845–0.848 on validation/test after feature enrichment).
- ASIN mapping to convert internal node IDs to real Amazon ASINs for frontend consumption.
- Production-capable FastAPI backend implementing a /recommend endpoint with JWT/HTTPBearer protection and environment-based configuration.
- Saved artifacts to support reproducibility.

Fellow (DL/ML Track), Buildables (formerly Bytewise)

Aug 2025 – Nov 2025

- Selected for a competitive 12-week mentor-led fellowship focused on Deep Learning and Machine Learning.
- Conduct hands-on project work covering data preprocessing, model prototyping, training, validation, and iterative optimization following industry best practices.
- Collaborate directly with industry track leads and mentors through weekly code reviews.
- Completed short projects during the fellowship, including:
 - ANN vs CNN comparison on the MNIST dataset (model design, training, and evaluation).
 - Cat vs Dog classifier (transfer-learning pipeline with data augmentation and evaluation).
- Final project: **Agentic RAG-based Smart Farming Advisor:**
 - Tabular crop predictor (XGBoost / RandomForest) with preprocessing (scaling, encoding, feature selection) and saved model artifact (models/croppredictor.pkl).
 - Disease detector (transfer-learning EfficientNet CNN) for leaf-image classification; produced inference pipeline and saved model artifact (disease_efficientnet_final.h5).
 - RAG pipeline: document chunking, embedding generation, FAISS index creation, and a generator fallback for low-recall cases.
 - Simple rule-based router to dispatch queries to crop predictor, disease detector, or retrieval module; implemented evaluation scripts and reported metrics (crop predictor test acc 0.86; disease detector test acc 0.92; RAG hit-rate@5 0.78; routing accuracy 0.88).
 - Delivered a reproducible repository with code, notebooks, model artifacts, FAISS index, evaluation scripts and a final project report.

Projects

Feedback Flow (Final Year Project)

2024 – 2025

- Designed Flask APIs for integration between frontend app with Sentiment Analysis Bert model which i

trained and fine tuned on Uber review dataset .

- Implemented MYSQL database schema and used Rasa framework to train and build Chatbot for the NUML transport app that fetches information directly from the database.
- Built a FastAPI proxy that authenticates (x-api-key), forwards Flutter messages to Rasa's REST webhook, and returns a single "reply":"..." response for the app.
- Exposed the local proxy with ngrok for cross-network testing and documented the runbook (ports, start order, headers) to enable Flutter Rasa end-to-end integration.
- Developed interactive visualizations in the Flutter app to present chatbot responses, sentiment results, and basic analytics.

WhatsApp Automation for Small Restaurants — n8n

- Built an end-to-end WhatsApp automation workflow using n8n, leveraging the Gemini API as the LLM and Google Sheets as the datastore.
- Implemented an AI agent with simple memory for context-aware order intake, templated confirmations, and status updates over WhatsApp.
- Deliverables: runnable n8n workflows and test artifacts demonstrating end-to-end functionality.

Academic and Personal Projects

- Customer churn prediction using ANN.
- Data visualization using Data Cubes.
- Spam email detection using Naive Bayes.
- Titanic survival prediction in Python.
- Amazon customer review sentiment analysis.

Education

BS in Artificial Intelligence

2022 – 2026

National University of Modern Languages (NUML), Islamabad

FSC (Pre-Engineering)

2019 – 2021

Askaria College, Rawalpindi

Major: Physics, Chemistry, Mathematics

Matriculation

2017 – 2019

F.G Sir Syed Boys Secondary School, Rawalpindi

Major: Biology, Chemistry, Physics

Certifications

- Buildables Fellowship (Deep Learning/Machine Learning Track) – Completion Certificate
- Graphic Designing (CorelDRAW, Photoshop, Illustrator, InPage) – Axiom Institute, Rawalpindi
- Data Science Foundations – Online MOOCs